M2 DATA SCIENCE PROJECT SPECIFICATION

Project Specification — Clinical Risk Modeling & Explainability

Domain: Tabular Medical Records & Automated Triage • **Scope:** Portfolio CV Project / Machine Learning

1. Context & Objective

In hospital management, early identification of patients at high risk of adverse events (such as 30-day unplanned readmissions or acute cardiac failure) enables targeted care and optimal resource distribution. The objective of this project is to build a reliable predictive machine learning system on tabular clinical data without requiring medical domain expertise, paired with transparent feature attribution (explainability) essential for medical decision support.

2. Dataset Specifications & Technical Scope

RECOMMENDED DATASET

Heart Failure Prediction Dataset (Kaggle) or **Diabetes 130-US Hospitals** (UCI ML Repository). Clean, structured tabular `.csv` format.

FEATURE CHARACTERISTICS

Continuous and categorical variables: age, resting blood pressure, serum cholesterol, fasting blood sugar, max heart rate, and patient medical history.

PREDICTION TARGET

Binary target variable `Target ∈ {0, 1}` (0 = stable/not readmitted, 1 = high risk/readmitted within 30 days).

HARDWARE CONSTRAINTS

Moderate volume (~1,000 to 100,000 rows). Entirely runnable on a standard laptop CPU using Python 3.10+ in under 5 minutes.

3. Project Phases & Deliverables

Phase	Algorithmic Tasks	Libraries & Tools	Deliverable
01. Data Prep	Missing value imputation, categorical encoding (Target/One-Hot), scaling, stratified 80/20 train/test split.	`pandas`, `scikit-learn`	Preprocessing Pipeline
02. Modeling	Baseline comparison (Logistic Regression vs. LightGBM / XGBoost), hyperparameter tuning.	`lightgbm`, `optuna`	Serialized Model (`.pkl`)
03. Explainability	Compute global and local SHAP values (Summary Beeswarm Plot, individual Waterfall Plots per patient).	`shap`, `matplotlib`	SHAP Impact Reports
04. Deployment	Lightweight REST microservice returning the predicted risk score and the top 3 driving features.	`fastapi`, `uvicorn`	Containerized API `/predict`

4. Evaluation Metrics & Validation Criteria

In healthcare applications, false negatives carry significantly more risk than false positives. Raw accuracy is strictly misleading due to class imbalance. The model must satisfy the following validation thresholds:

Precision-Recall AUC (PR-AUC) & ROC-AUC: Minimum target ROC-AUC ≥ 0.85. PR-AUC prioritized to accurately measure performance on minority positive outcomes.

Probability Calibration: Target Brier Score ≤ 0.15. Apply `CalibratedClassifierCV` (Platt Scaling) so predicted confidence accurately reflects empirical clinical event frequencies.

Inference Latency: Single-sample inference latency ≤ 15 ms via the REST endpoint.

5. Inference Microservice Architecture (FastAPI)

The serving API will expose a production-ready endpoint accepting patient feature payloads in JSON:

- **Endpoint:** `POST /api/v1/patient-risk`
- **Input Schema:** Demographic data, lab measures, and history validated with `Pydantic` models.
- **Structured Response:** `{ "risk_score": 0.74, "risk_level": "HIGH", "top_drivers": ["cholesterol", "age", "resting_bp"] }`
- **Reproducibility:** Pinned `requirements.txt` and standalone `Dockerfile` for one-step local or cloud deployment.

6. CV Framing & Resume Highlights (M2 Data Science)

Recommended Title: *Clinical Risk Stratification Engine & Explainability Microservice (LightGBM • SHAP • FastAPI)*

Resume Bullet Points: Engineered a calibrated gradient-boosted tree (LightGBM) on 10,000+ patient records for 30-day readmission prediction, achieving 0.88 ROC-AUC and 0.12 Brier Score; integrated SHAP tree explainers to produce real-time patient-level feature attributions; deployed a containerized FastAPI endpoint with sub-15ms inference latency.